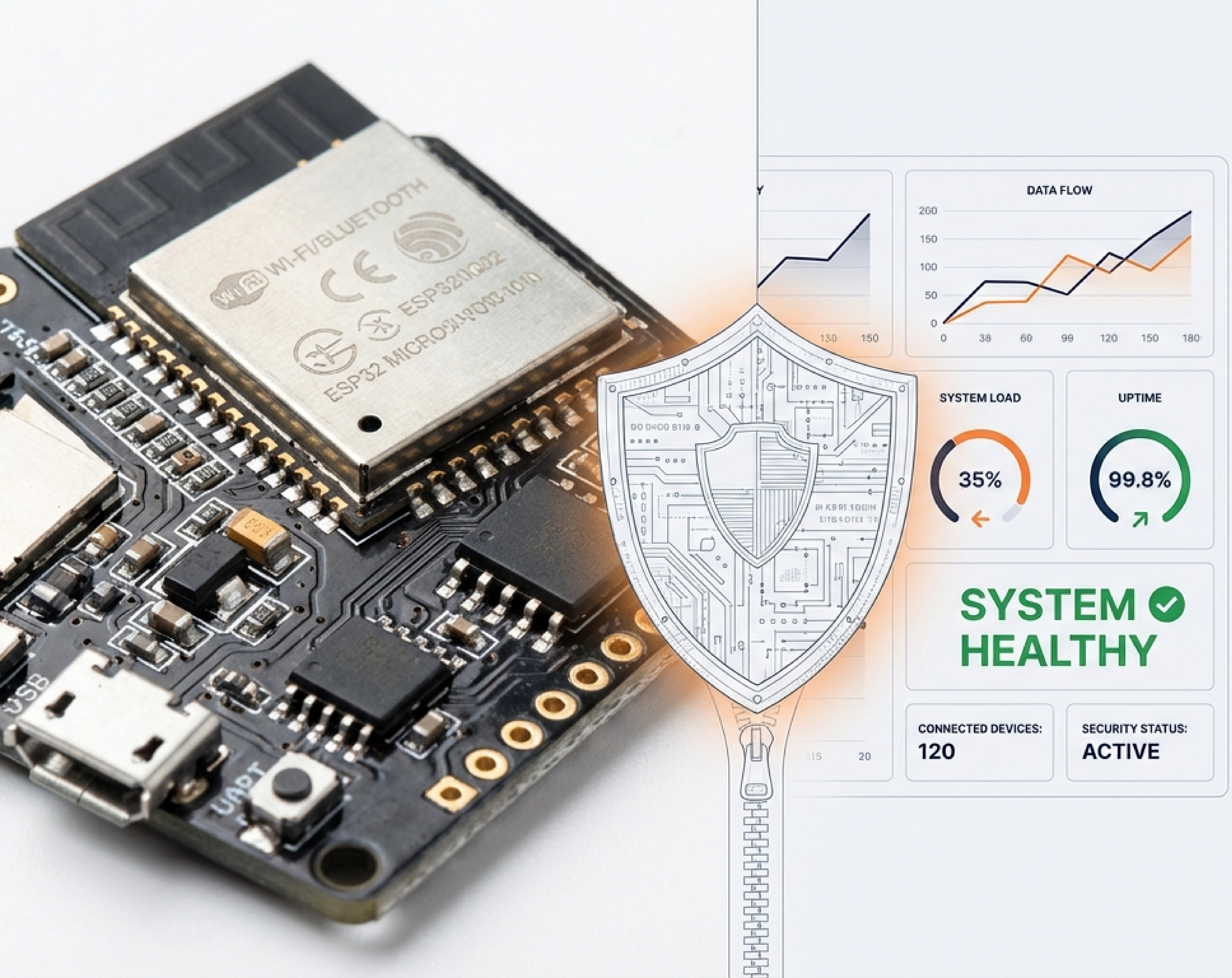


SentinelOTA

Bridging Embedded Constraints and Lifecycle Security

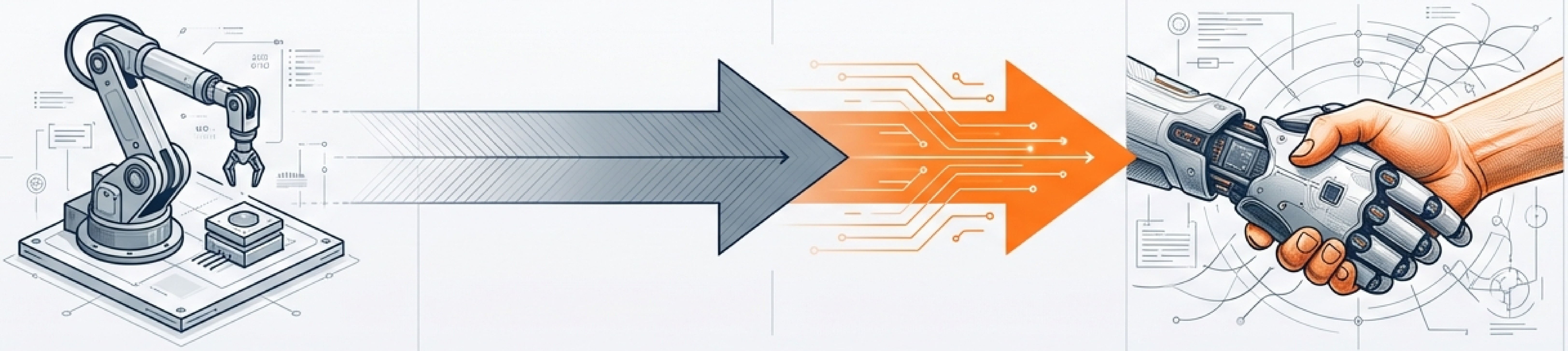
A Unified, Resilient Framework for the Industry 5.0 Heterogeneous IoT Ecosystem.

Project by: [Aspiring Offensive Security Engineer Name]
Context: Cyber Security & Offensive Engineering

The Industry 5.0 Imperative

From Connected to Resilient.

Industry 4.0 | **Industry 5.0**



Efficiency
Connectivity
Fragile

In Industry 5.0, a system that cannot heal itself is a liability.

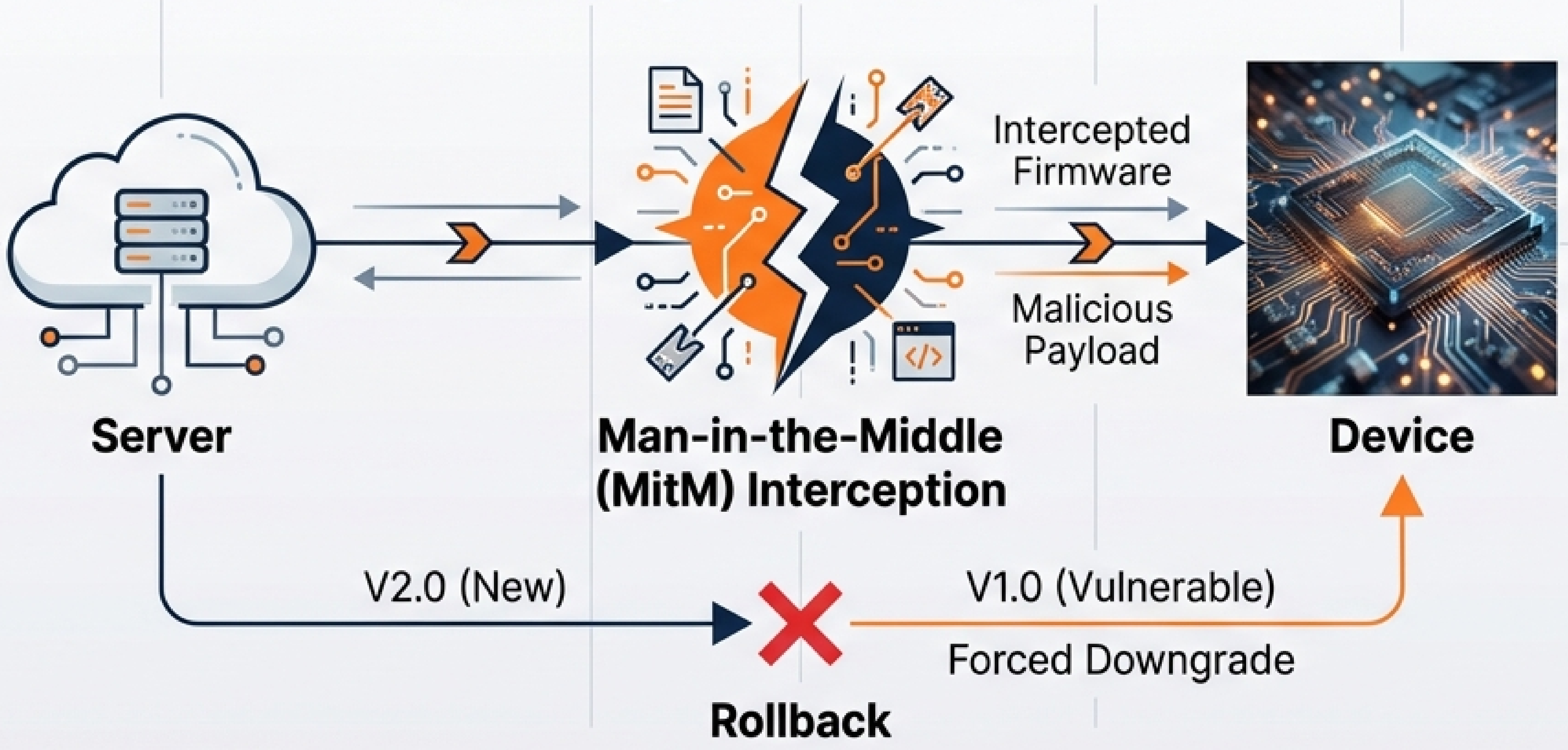
Resilience
Human-Centric
Sustainable

Context: As factory floors rely on mixed fleets of devices, the risk shifts from simple downtime to human safety. Resilience means automatically recovering from compromise, not just preventing it.

The Attack Surface: Updates as the Primary Vector

Viewing the update cycle through an Offensive Security lens.

Anatomy of an OTA Attack



The Deploy-and-Forget Risk: Current systems confirm code delivery but fail to verify post- post-deployment behaviour.

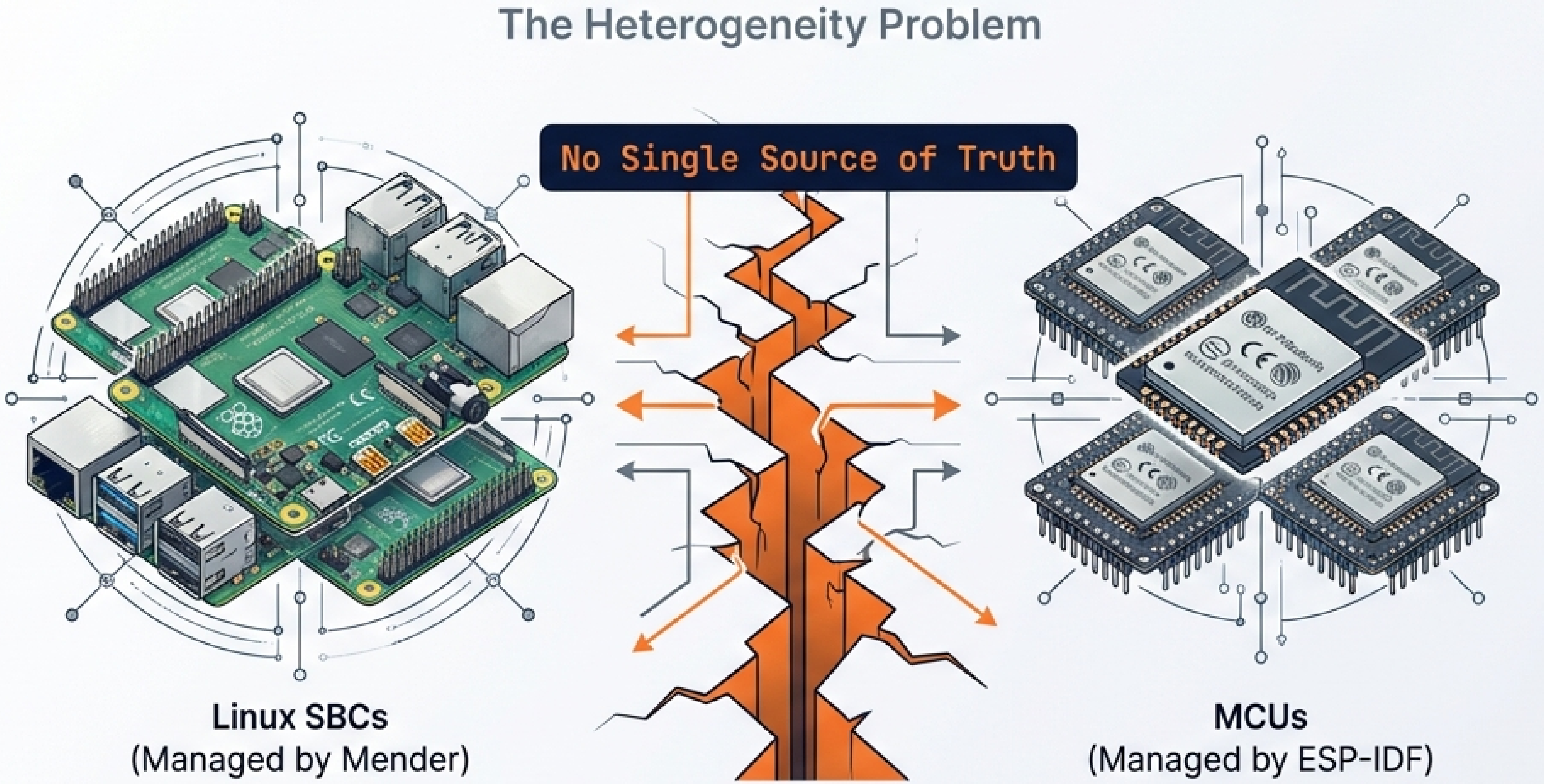
Top Threats:

1. Man-in-the-Middle (MitM) attacks intercepting firmware.
2. Rollback attacks forcing devices to older, vulnerable versions.

“I viewed the update cycle not as a feature, but as the largest gap in the security perimeter.”

Fragmented Architectures Create Security Blind Spots

The Heterogeneity Problem: Viewing the device fleet as disconnected islands.



Mixed device fleets lack a unified framework to handle mutual authentication across different architectures without massive overhead.

Resource Constraints: Heavy encryption (RSA) drains ESP32 batteries, requiring a tailored approach.

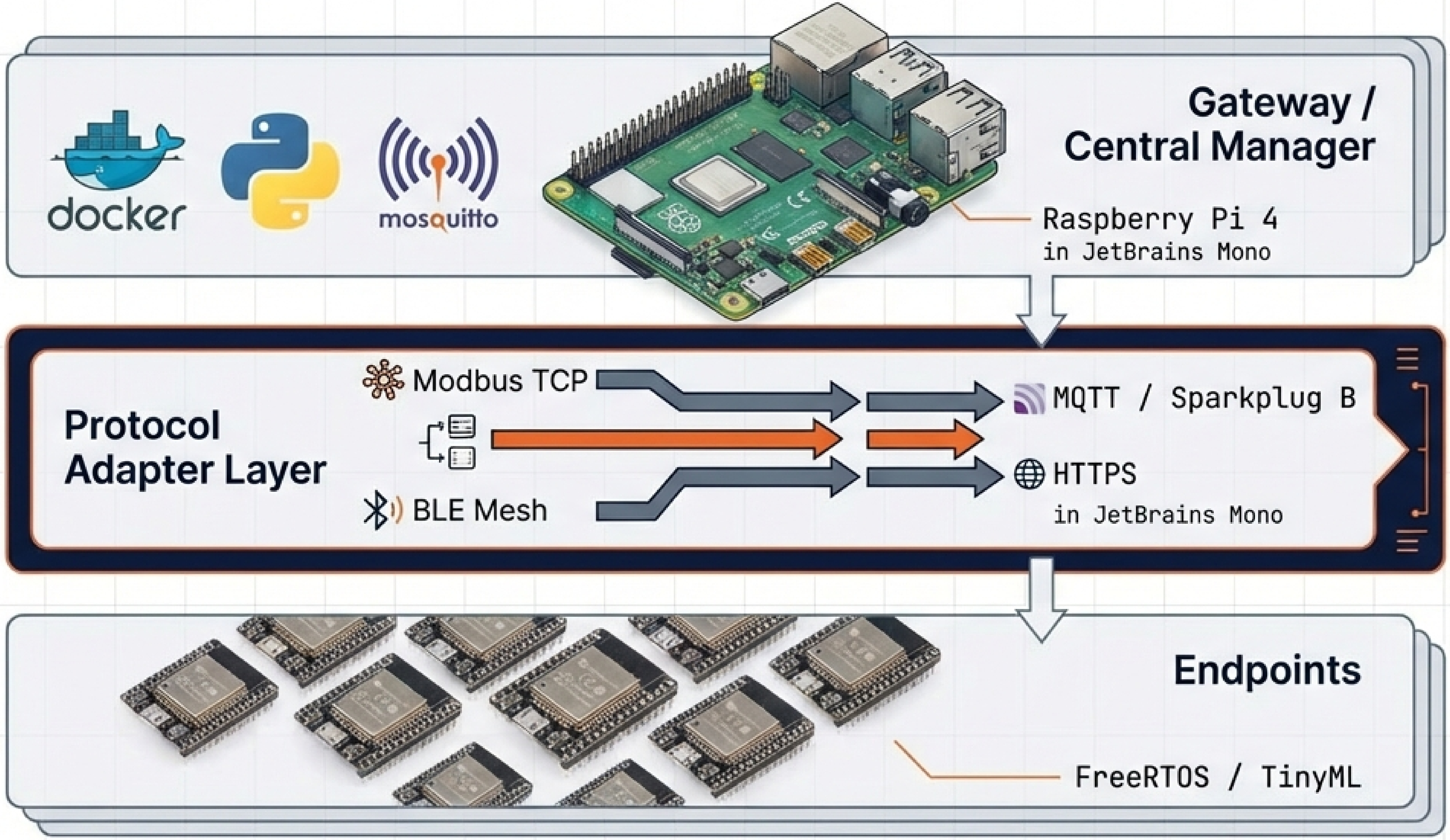
Context: Heterogeneity isn't just a management headache; it's a foundational security vulnerability that attackers exploit.

Introducing SentinelOTA

A resilient lifecycle framework designed for the edge **constraints of Industry 5.0.**

NotebookLM

Architecture: From Central Manager to Edge Sensor

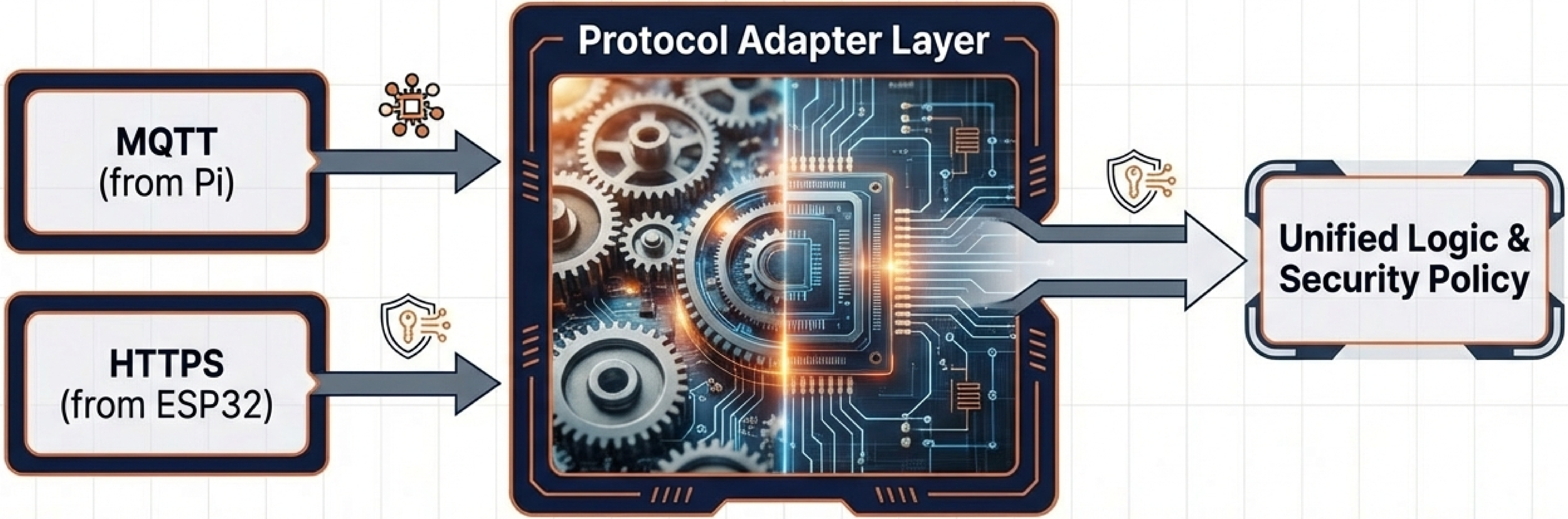


1. **Trust:** Certificate Authority (CA) and NVS Storage.
2. **Delivery:** Binary signing and distribution via MQTT/HTTPS.
3. **Defense:** Local TinyML baseiines vs. Global analysis via Scikit-Learn.

The Protocol Adapter Layer

This translation layer allows a single gateway to manage **security certificates** and **firmware signatures** for both MCUs and SBCs simultaneously.

Benefit: Eliminates the need for multiple management dashboards, closing the gap between Linux and FreeRTOS environments.



Security First: Post-Quantum Readiness & Mutual TLS

PQC-Ready Signing

Utilising JetBrains Mono for **liboqs** to implement **JetBrains Mono for Kyber** and **JetBrains Mono for Dilithium** algorithms. Securing against future quantum decryption capabilities.

mTLS (Mutual TLS)

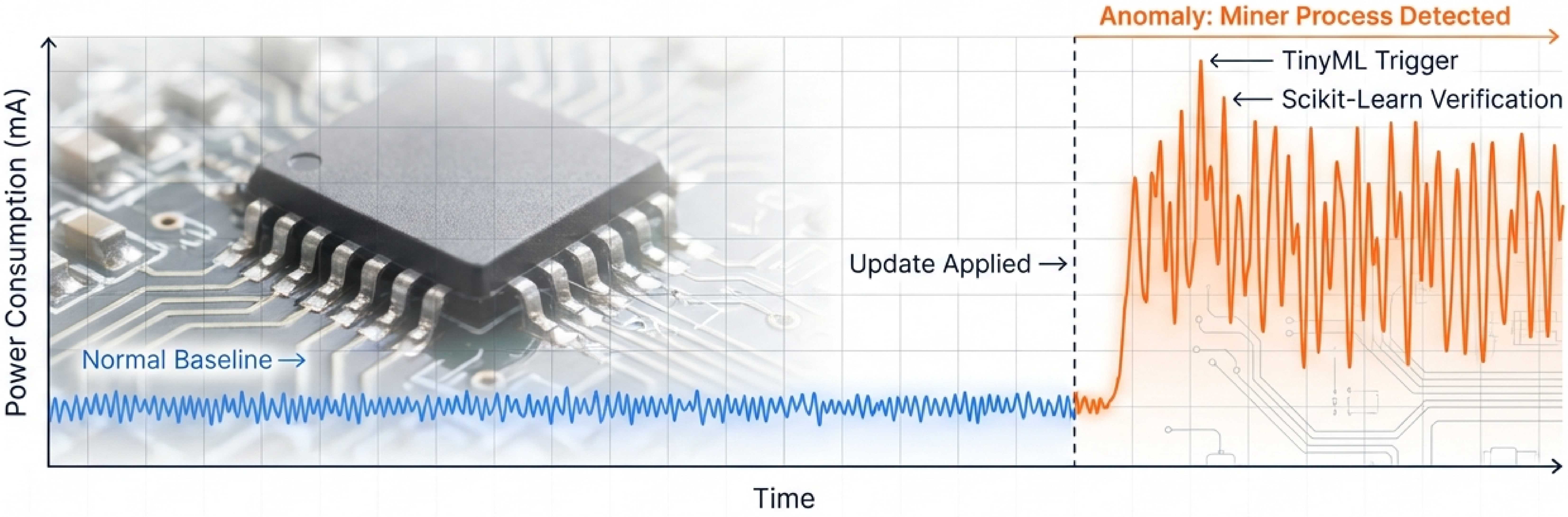
Every handshake is verified. The server authenticates the device, and the device authenticates the server.

The Offensive Hook

Standard JetBrains Mono for TLS is vulnerable to sophisticated **JetBrains Mono for MitM** attacks; **JetBrains Mono for mTLS** ensures identity verification at both ends.

The Digital Immune System

Post-Update Behavioural Monitoring

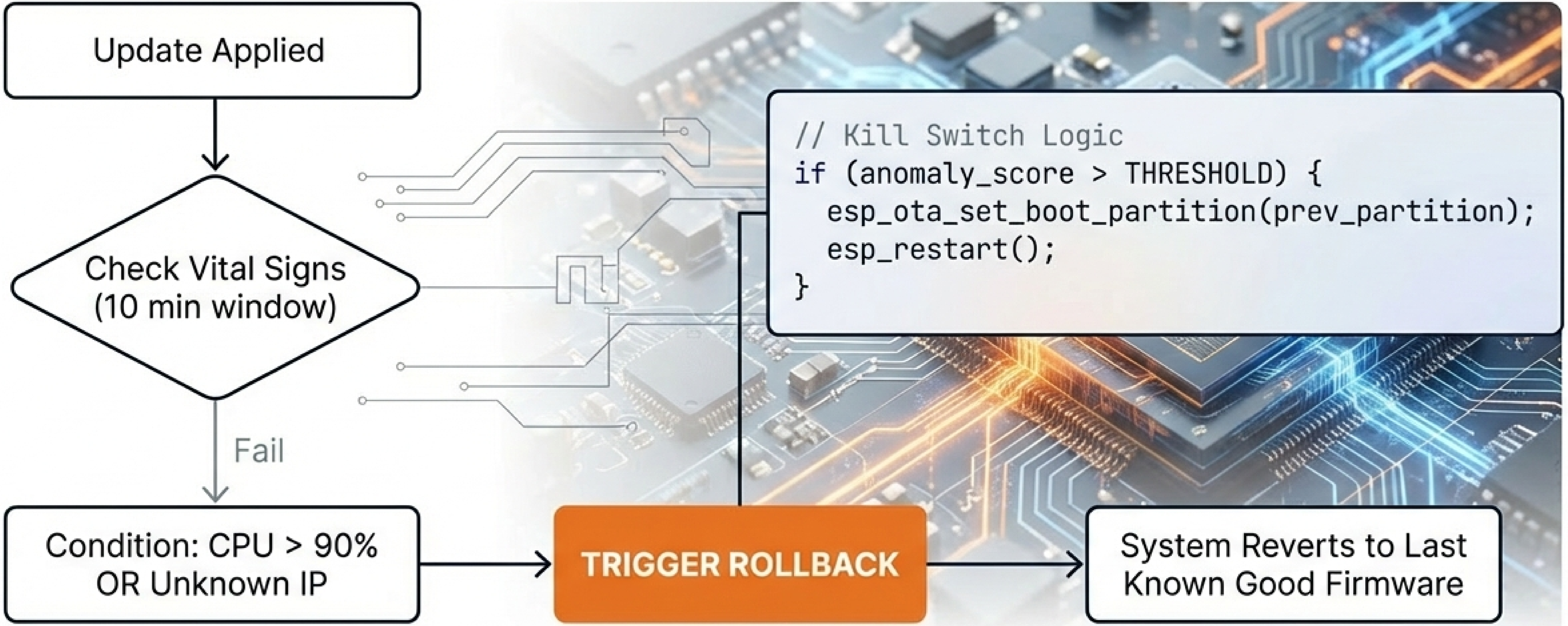


We do not trust the code just because it is signed. We verify its behaviour in the real world.

Mechanism: TinyML creates local baselines for expected power usage and network traffic. Deviation triggers an immediate response.

Automated Zero-Touch Recovery

Ensures the device remains operational even if the deployed firmware is malicious or buggy.



Implementation Stack

Hardware

- Raspberry Pi 4 (Gateway)
- ESP32 (Endpoint)

Software & Backend

- Python / Flask
- Docker Containers
- Mosquitto (MQTT Broker)

Firmware & Edge

- FreeRTOS
- ESP-IDF
- TinyML Lite

Security & VAPT

- liboqs (Post-Quantum Crypto)
- Parrot OS Tools (Stress Testing)

Built with industry-standard, scalable tools to ensure real-world applicability.

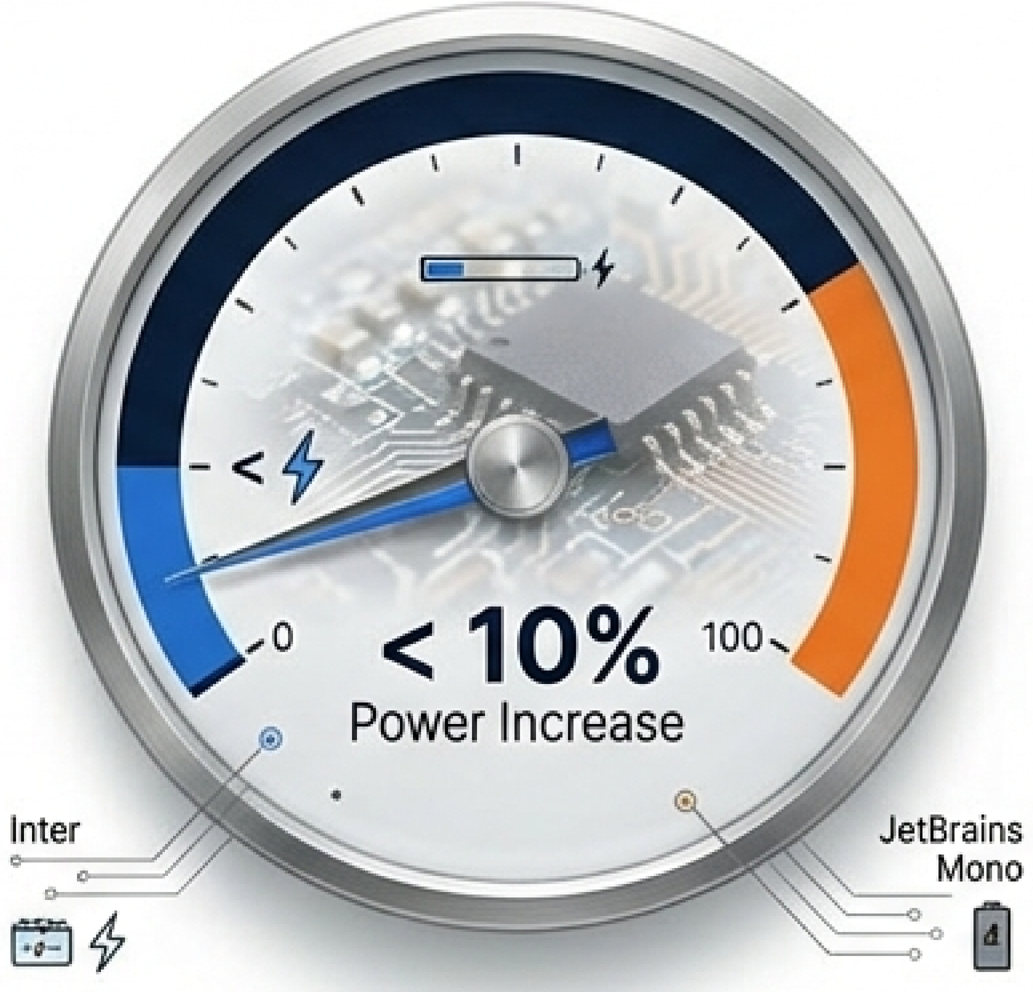
Validating Resilience: Evaluation Metrics

The system adds high security with minimal resource penalty.

Security Success

100% Block Rate
(Replay Attacks & Unauthorized Uploads)

Performance Overhead



< 10% Power Increase
(During Monitoring Phase)

Reliability

100% Auto-Rollback Success
(Simulated Malicious Update)

Technical Swiss Editorial

Roadmap & Future Scope

Scaling SentinelOTA from a single factory floor to a global fleet.

Current Status

- Functional Prototype
- Local Anomaly Detection
- mTLS Enforced.

Phase 2 Integration

- Enterprise SIEM Integration
- Dashboarding.

Future Scope

- Federated Learning
- Fleet-wide immunity sharing.

From Smart to Secure & Resilient

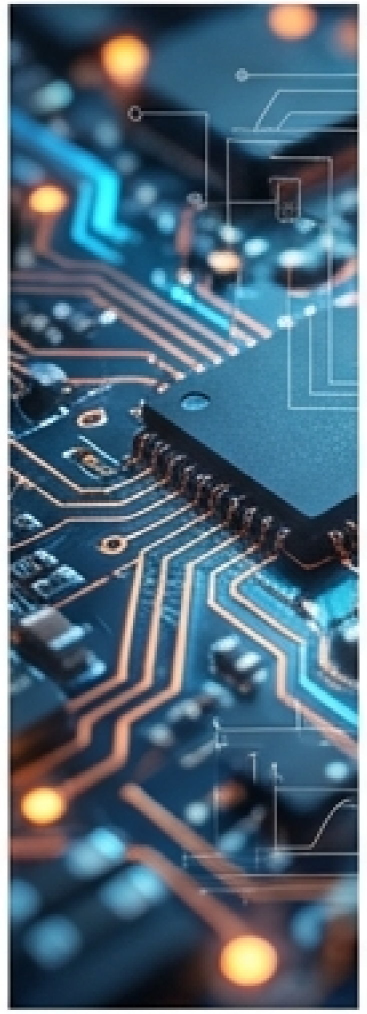
SentinelOTA successfully closes the gap between deployment and safety. By combining PQC-signing with behavioural analytics, we ensure Industry 5.0 systems can withstand the threats of tomorrow.

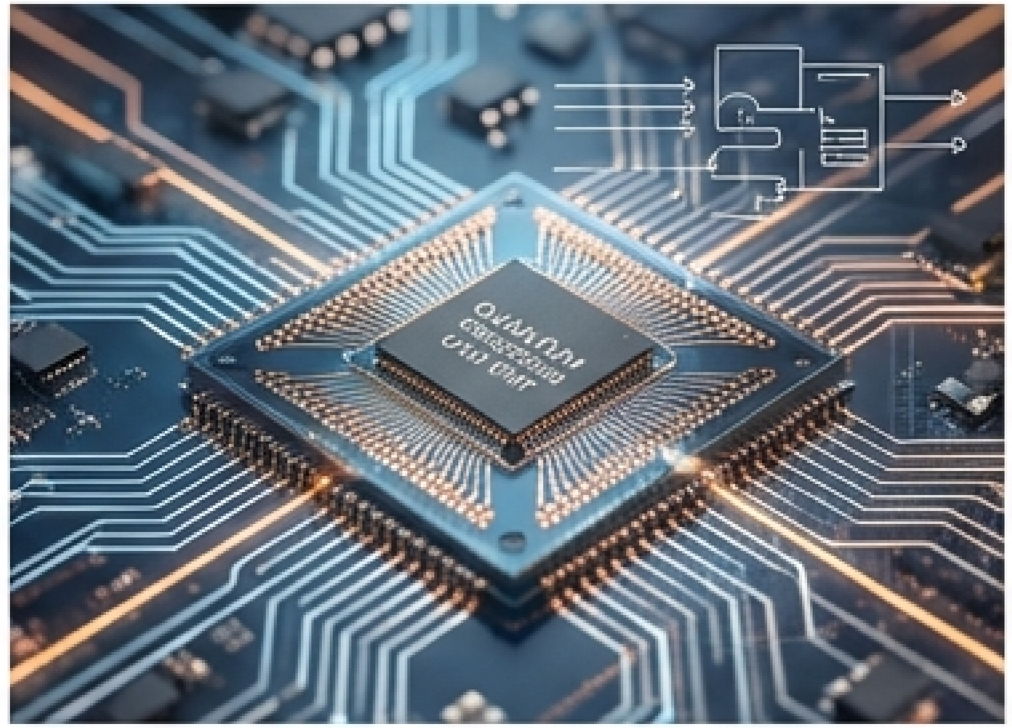
Security is not a state; it is a lifecycle.

References & Resources

Key Academic References

- Industry 5.0: Towards a Resilient European Industry (European Commission)
- Post-Quantum Cryptography for IoT: Challenges and Opportunities (IEEE)
- Anomaly Detection in Embedded Systems using TinyML





Project Repository

Codebase and Documentation:

`github.com/[username]/sentinelota` (Proof of Concept)

Contact

- [Presenter Name]
- Aspiring Offensive Security Engineer
- [Email Address]